

CE 3111

STRUCTURAL ANALYSIS AND DESIGN - I

SECTION B CLASS NOTE

SET 5-8 | Concept + Formula + Bangla Explanation + Shortcut + Correct Sketch + QB Match

SET 5	Influence Line Diagram (ILD)
SET 6	Approximate Frame Analysis: Portal + Cantilever + Factor
SET 7	Mill Bent / Portal Application + Space Truss
SET 8	Three-Moment Equation + Suspension Cable

NOTE SCOPE This follows the same set-distribution convention as the previously prepared CE 3111 Question Bank Analysis: Section B = Set 5-8. Some old papers/backlog print the two section letters in reverse order, so identify the block by topic, not by the old printed letter.

Source basis: uploaded Ramjan Sir class note/sheet, ILD assignment, 2016-2024 Question Bank, 2024 Backlog, and topic-wise solved QB files for ILD, cantilever method and three-moment equation; cable equations are aligned with the uploaded cable note.

Made by 2223033

How to Use This Note

Section B-তে 4 সেটের মধ্যে 3 সেট answer করতে হবে। তাই note-টি চারটি complete set ধরে সাজানো হয়েছে। প্রতিটি set-এর ভেতরে concept, formula, solving workflow, Bangla explanation, memory shortcut, exam trap এবং matching Question Bank pattern দেওয়া হয়েছে।

Set	Core topics	Exam nature	Suggested role
5	Influence Line Diagram	Beam/frame response under moving unit load	Core
6	Portal + Cantilever + Factor	Approximate frame analysis	Core; recent portal+cantilever trend
7	Mill Bent + Space Truss	Portal application + 3D equilibrium	Backup/Core if 3D is strong
8	Three-Moment + Cable	Continuous beam + cable statics	Core; formulaic

3-of-4 STRATEGY Default route: Set 5 + Set 8 + Set 6. Keep Set 7 as the switch/backup. If 3D space truss is comfortable for you, Set 7 can replace the set you find hardest.

Quick Memory Map

Topic	Memory shortcut
ILD	$x \rightarrow$ piecewise equilibrium $\rightarrow$ ordinates $\rightarrow$ join; every support/hinge creates a new range
Müller-Breslau	Release the required response i mpose +1 displacement/rotation $\rightarrow$ deflected shape gives sign
Portal	MID-MID + I = 2E: inflection at beam/column middle; interior column shear twice exterior
Cantilever	MID-MID + $P \propto A \cdot x$; first find centroid of column group
Factor / vertical-load backup	Beam axial ≈ 0 ; beam POI $\approx 0.1L$ from ends; distribute joint moment by $K=EI/L$
Space Truss	XYZ first, then direction cosines; assume tension and let sign decide T/C
Mill Bent	Use its own assumptions; do not blindly apply building-portal 2:1 shear rule
Three-Moment	3 supports $\rightarrow$ 2 spans $\rightarrow$ 1 equation; then solve spans separately
Cable	Vertical reactions like a simply supported beam + constant H; $T = \sqrt{(H^2 + V^2)}$

Contents

SET 5 - Influence Line Diagram (ILD)

SET 6 - Approximate Frame Analysis: Portal + Cantilever + Factor

SET 7 - Mill Bent / Portal Application + Space Truss

SET 8 - Three-Moment Equation + Suspension Cable

Section B Theory / Derivation List + Final Formula Sheet + 3-of-4 Checklist

SET 5 - Influence Line Diagram (ILD)

QB STATUS Annual-core family. Beam ILD is the oldest recurring pattern; recent papers repeatedly add a frame ILD. Theory on straight-line ILD / importance also appears.

5.1 What is an Influence Line?

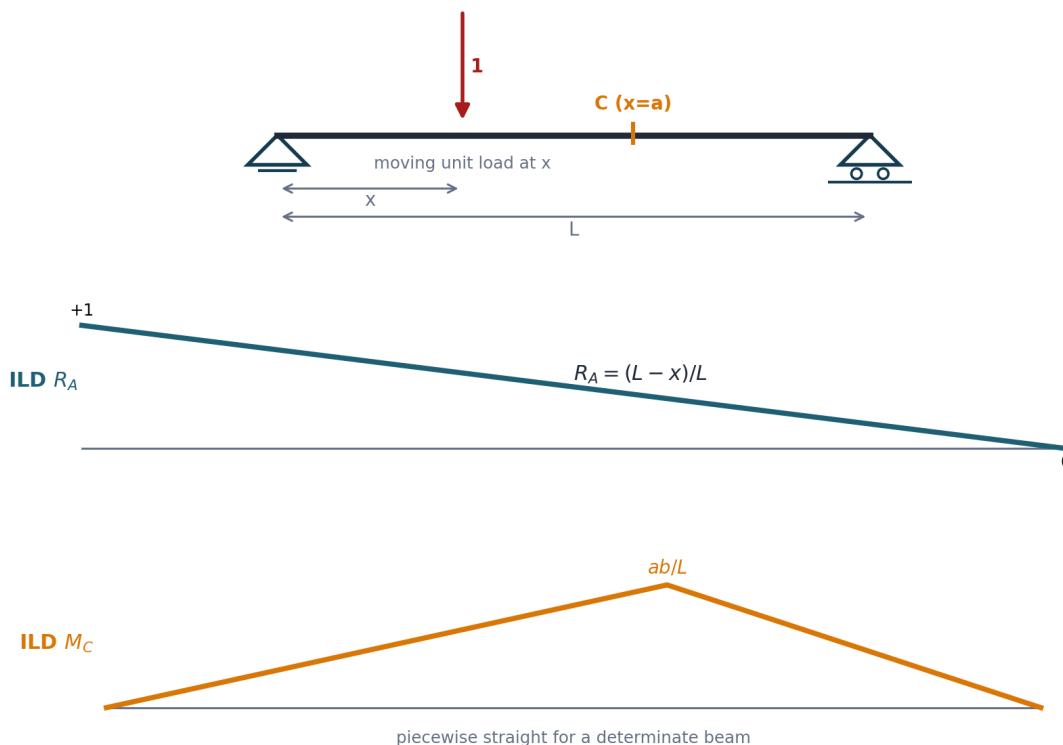
Influence Line (প্রভাব রেখা) হলো এমন graph যা structure-এর একটি নির্দিষ্ট response - যেমন reaction, shear বা bending moment - কীভাবে পরিবর্তিত হয় তা দেখায়, যখন একটি downward unit load structure-এর উপর এক position থেকে অন্য position-এ move করে।

Term	Meaning
Response point	যে reaction / section shear / moment-এর ILD আঁকছি, সেই location fixed থাকে.
Moving load position x	Unit load-এর position পরিবর্তিত হয়.
ILD ordinate	এই x-position-এ fixed response-এর value.
Positive / negative region	Moving load এই region-এ থাকলে response-এর sign কী হবে তা দেখায়.

সবচেয়ে জরুরি পার্থক্য SFD/BMD একটি fixed loading-এর জন্য member-এর along-length response দেখায়; ILD একটি fixed response location-এর জন্য moving load-এর position পরিবর্তনের effect দেখায়.

SFD/BMD	Influence Line
Loads fixed; section moves conceptually along member	Response section fixed; unit load moves
Shows V/M along structure for one load case	Shows one response as function of load position
Used for ordinary static loading	Essential for moving loads: bridge, crane, wheel loads

Influence line: one response at a fixed point while a unit load moves



5.2 Equilibrium Method - The Main Exam Method

Ramjan Sir-এর class note-এ ILD মূলত equilibrium method দিয়ে করা হয়েছে। Trick হলো পুরো structure-এর জন্য একটাই expression না লিখে x-এর position অনুযায়ী range ভাগ করা।

1. Draw the structure and let a downward unit load = 1 act at distance x from a chosen origin.
2. Mark every support, internal hinge and target section. These points usually divide the x-ranges.
3. For each x-range, write independent equilibrium equations and obtain the required reaction/shear/moment as a function of x.
4. At each critical point (support, hinge, section, end), substitute x to get ordinates.
5. Plot ordinates with correct sign and join by straight lines for a statically determinate member segment.
6. Check: reactions should satisfy $\sum V=1$; internal hinge moment must be zero; shear ILD has a unit jump at the section itself.

Bangla shortcut “Range না ভাগ করলে ILD ভুল হওয়ার chance সবচেয়ে বেশি।” Support/hinge/target section-এর আগে-পরে x-এর equation বদলাতে পারে.

5.3 Simply Supported Beam - Must-Know Formula

A-B simply supported beam-এর span L. Unit load A থেকে x দূরে। C section A থেকে a দূরে; $b=L-a$.

Support reactions

$$R_A = (L - x) / L$$

$$R_B = x / L$$

Shear at C - using the lecture-style left-segment sign convention

$$V_C = -x/L, \quad 0 < x < a$$

$$V_C = (L-x)/L, \quad a < x < L$$

Bending moment at C

$$M_C = x b / L, \quad 0 < x < a$$

$$M_C = a (L-x) / L, \quad a < x < L$$

$$\text{Peak ordinate at C} = a b / L$$

Shear jump rule Unit load C-এর left side থেকে right side cross করলে shear ILD ordinate +1 jump করে: $V(C+) - V(C-) = 1$. এই check খুব useful.

5.4 Why is the ILD Straight for a Determinate Structure?

একটি determinate structure-এর নির্দিষ্ট x-range-এ equilibrium equation থেকে response সাধারণত x-এর first-degree linear function হয়। তাই ঐ range-এর ILD straight line হয়। Internal hinge/support-এর কারণে নতুন range শুরু হলে slope বদলাতে পারে; পুরো ILD একটি single straight line নাও হতে পারে, কিন্তু প্রতিটি determinate segment piecewise straight হয়।

Exam-ready one-line answer For a statically determinate structure, the response can be obtained solely from equilibrium and is a linear function of the moving unit-load position within each structural segment; hence the influence line is piecewise straight.

5.5 Internal Hinge / Overhang / Multi-Support Beam

Old and recent QB-তে internal hinge এবং overhang খুব common. এখানে hinge হলো extra equilibrium advantage: hinge moment = 0 ব্যবহার করে reaction relation পাওয়া যায়।

Internal hinge	Use $M_{\text{hinge}}=0$ for the appropriate released part; x crossing hinge usually changes equation range.
Overhang	Keep unit load on overhang in its own x-range; reactions may become negative.
Multiple supports in a determinate assembly	Use local hinge equations + whole-structure equilibrium; do not treat it as a simple beam.
Shear at a point	Write left-of-section and right-of-section expressions separately; preserve the unit jump.
Moment at a point	Take moment of the convenient side; ILD is continuous at ordinary section unless a response release changes it.

5.6 Frame ILD - Recent High Priority

Frame ILD-তে moving load সাধারণত specified horizontal beam/girder-এর উপর move করে, কিন্তু response হতে পারে support reaction, column shear বা joint/section moment. Beam-এর unit load frame-এর পুরো reaction system change করে।

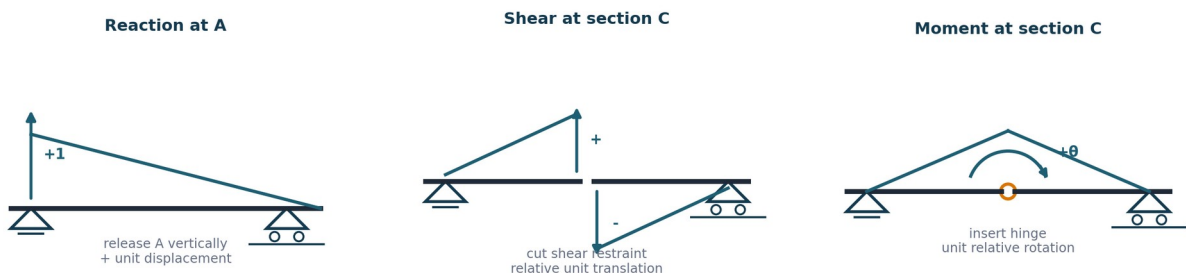
- Fix axes and the path of the moving unit load. Do not move the load on members not specified by the question.
- Set x along the loaded beam and solve global reactions first.
- Use internal hinge / joint equilibrium when available.
- For shear or moment at a frame point, cut the convenient part and write the response as a function of x.
- Create new x-range whenever the unit load crosses a support, hinge, column line or target section.

Frame shortcut Beam-এর মতো “one span formula” blindly ব্যবহার করো না। Frame geometry থেকে moment arm বদলে যায়; always draw FBD for the selected x-range.

5.7 Müller-Breslau Principle

Müller-Breslau principle হলো ILD-এর qualitative shape বোঝার দ্রুত method। যে response-এর ILD চাই, সেই response-এর corresponding restraint release করে positive direction-এ unit displacement বা unit rotation দিলে released structure যে displaced shape নেয়, সেটি ঐ response-এর influence line-এর shape-এর proportional।

Müller-Breslau principle: release the response and impose a positive unit displacement/rotation



Reaction → release translation restraint; shear → relative translation; moment → insert hinge and impose relative rotation.

Wanted response	Release	Impose positive action
Support reaction	Remove corresponding support restraint	Unit translation in positive reaction direction
Shear at a section	Cut section so relative vertical movement is possible	Positive relative unit translation
Moment at a section	Insert hinge / release moment restraint	Positive relative unit rotation

5.8 ILD Exam Technique - 60 Second Setup

Step	What you write on script
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1	Take moving unit load 1 at x from the stated origin.
2	Write x-ranges above the figure before solving.
3	Derive response function separately for every range.
4	Make a small ordinate table at A, B, hinge, section, C, ...
5	Plot sign first; then join straight lines.
6	Final checks: $\Sigma R=1$, $M_{\text{hinge}}=0$, shear jump=1.

5.9 Matching Questions from the Question Bank

Year	Matching question pattern (paraphrased)
2016	Two ILD problems: support reactions/shear, plus reaction/shear/moment with internal hinges.
2017	ILD for reactions at B, D, G and reaction moment at G.
2018	Beam ILD with required ordinates; ILD importance also appears in the paper.
2019	Frame ILD: support reaction and shear/moment at specified points.
2020*	Beam ILD for reactions at A/C/D and shear at points 1 and 2.
2021	Reactions at B/E/G + shear at C + moment at F.
2022	Detailed beam ILD ordinates and a separate frame ILD.
2023	Why ILD is straight + beam and frame ILD.
2024	Beam ILD + frame ILD with reactions/shear/BM.
2024 BL	Beam ILD + frame ILD again; same modern pattern.

Set 5 - Most Common Errors

- Using one equation across all x positions.
- Forgetting that the moving load is exactly 1.
- Missing the unit jump in shear ILD.
- Ignoring internal-hinge $M=0$ condition.
- Plotting a smooth curve in a determinate segment where the response is linear.
- Mixing SFD/BMD logic with ILD logic.

SET 6 - Approximate Frame Analysis: Portal + Cantilever + Factor

QB STATUS Annual-core family. 2022-2024 strongly favor Portal + Cantilever together. Factor/approximate vertical-load method is historically repeated and should be kept as a compact backup.

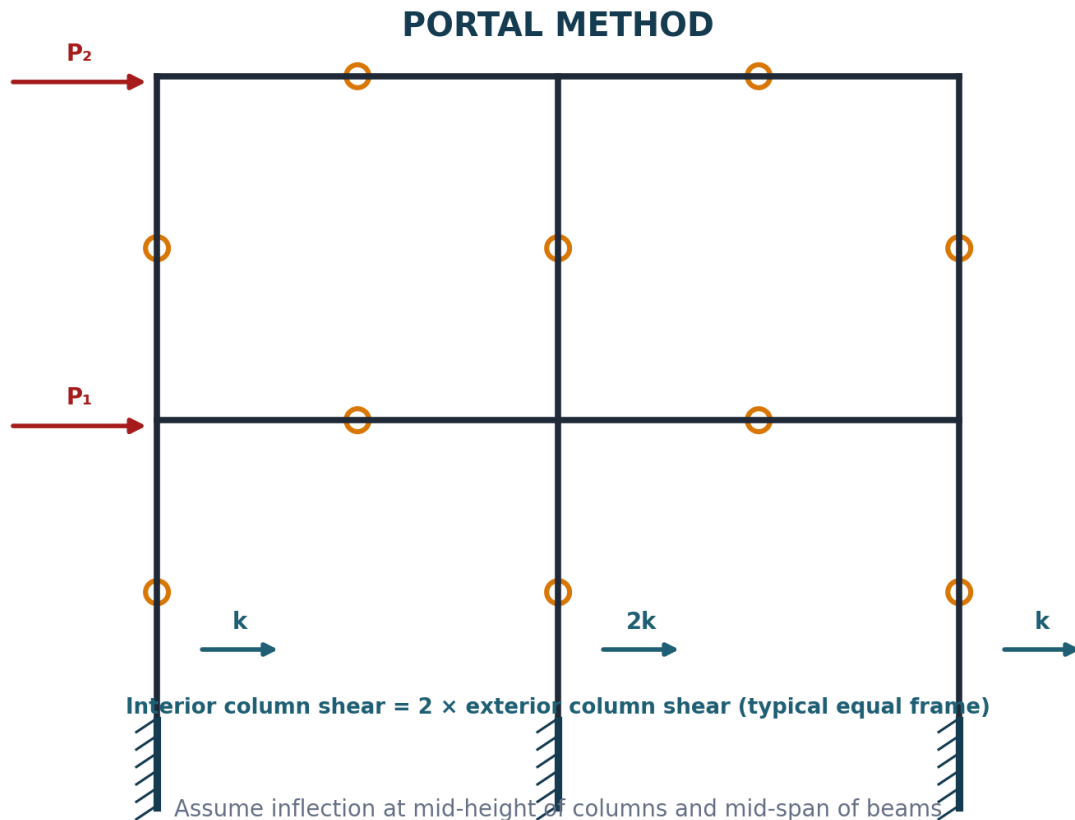
6.1 Why Approximate Analysis?

Multi-storey rigid frame সাধারণত highly statically indeterminate. Exact analysis-এ compatibility/deformation equations লাগে। Approximate analysis-এ realistic inflection-point ও force-distribution assumptions নিয়ে structure-কে equilibrium দিয়ে দ্রুত solve করা হয়।

Class-note principle First estimate the degree of static indeterminacy; then introduce an appropriate number of assumptions so that equilibrium becomes sufficient.

Method	Main physical idea	Key force distributed
Portal method	Frame behaves like a set of portals resisting storey shear	Column shear
Cantilever method	Whole frame behaves like a vertical cantilever resisting overturning moment	Column axial force
Factor / vertical-load backup	Loaded beams transfer unbalanced joint moments to columns by stiffness	Joint bending moment

6.2 Portal Method - Core Assumptions



Portal method assumptions used in the class note for building frames.

- Point of inflection is assumed at the center of each column and each girder/beam.

- At a storey, each interior column carries twice the horizontal shear of each exterior column for the usual equal-column frame idealization.

Memory trick PORTAL = "MID-MID + I=2E". MID-MID → column ও beam-এর middle-এ zero moment; I=2E → Interior column shear = 2 × Exterior.

6.3 Portal Method - Step-by-Step Numerical Workflow

- Compute storey shear V_s at each level = sum of all lateral loads at and above that level.
- Distribute V_s among columns. For a common frame with two exterior columns and n interior columns: if each exterior shear = k and each interior = $2k$, then $V_s = 2k + 2nk$.
- With inflection at column mid-height, obtain column end-moment magnitude from $M = V_c (h/2)$. Keep algebraic signs consistent.
- At every joint, enforce moment equilibrium to get beam-end moments. With a beam inflection at midspan, its moment diagram crosses zero at $L/2$.
- Get beam shear from member equilibrium: $V_b = (M_R - M_L)/L$ in algebraic form when no transverse load acts on that beam segment.
- Use vertical equilibrium of joints/columns to obtain column axial forces; accumulate effects downward storey by storey.
- Draw column and beam SFD/BMD after all end forces are tabulated.

Portal quick relations

Storey shear: $V_s = \Sigma$ lateral loads above the storey
 Typical portal split: exterior = k , interior = $2k$
 Column end moment magnitude: $M_c = V_c h / 2$

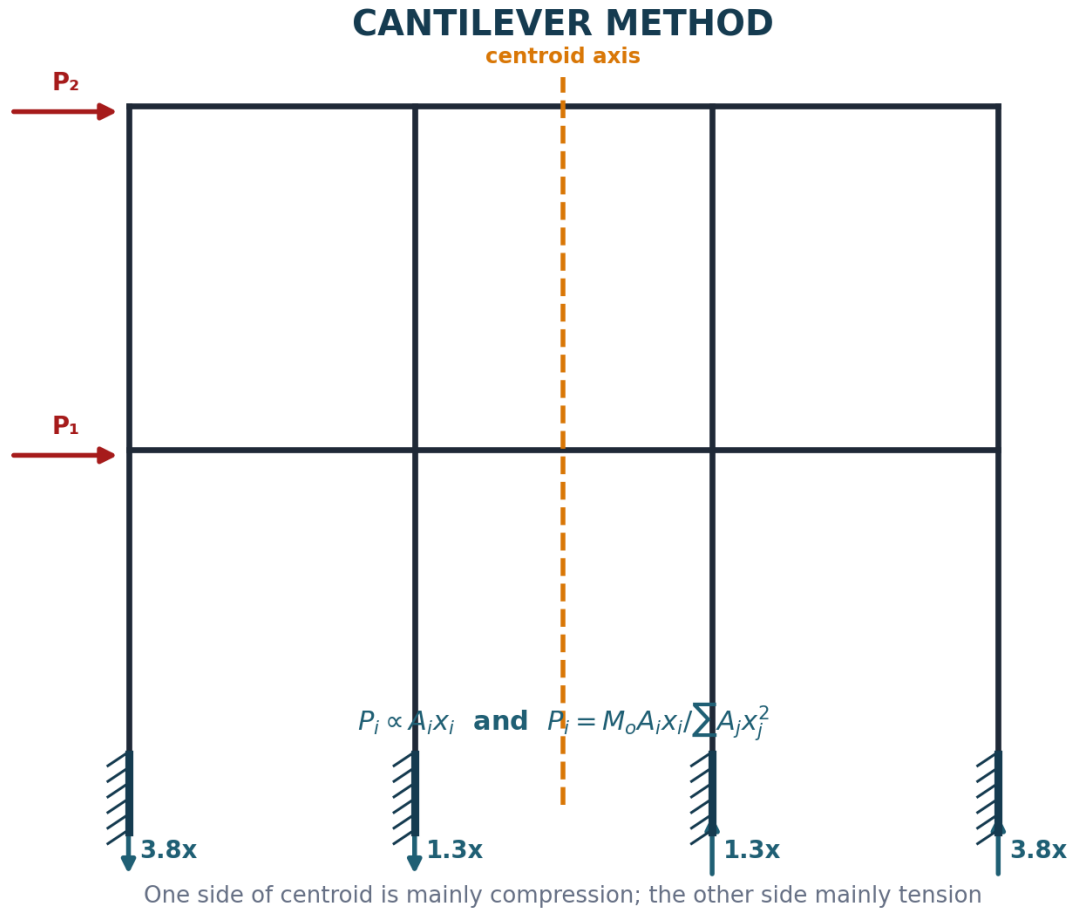
Important limitation The 2:1 interior/exterior shear rule belongs to the usual building-frame portal approximation. Do not transfer it automatically to the separate mill-bent method in Set 7.

6.4 Tiny Portal Check Example

এক storey-তে 3টি column (2 exterior + 1 interior) এবং total storey shear $V_s = 60$ kN হলে: $k + 2k + k = 60 \rightarrow k = 15$ kN. তাই column shears = 15, 30, 15 kN. Height $h = 4$ m হলে end-moment magnitudes = 30, 60, 30 kN·m respectively. এরপর joint equilibrium দিয়ে beam moments বের করবে।

QB MATCH - Portal Portal method / portal assumptions recur in 2016, 2019, 2021, 2022, 2023, 2024 and 2024 Backlog; the recent pattern usually asks complete member axial force, shear, moment and SFD/BMD.

6.5 Cantilever Method - Core Assumptions



The frame is treated like a vertical cantilever: overturning moment is resisted mainly by axial forces in columns.

- Point of inflection is assumed at the center of each column and girder.
- At any storey, axial force in a column is proportional to the horizontal distance of that column from the centroid (center of gravity) of the column group. The class solutions also include column area, so the general proportionality is $P_i \propto A_i x_i$.

Cantilever axial-force distribution

$$\begin{aligned}\bar{x} &= \sum(A_i x_i) / \sum A_i \\ P_i &= C A_i x_i \\ C &= M_o / \sum(A_j x_j^2) \\ \text{Therefore: } P_i &= M_o A_i x_i / \sum(A_j x_j^2)\end{aligned}$$

Here x_i is the signed horizontal distance from the centroidal axis and M_o is the overturning moment of the lateral loads above the storey cut. One side becomes mainly compression, the other side mainly tension.

Memory trick CANTILEVER = "MID-MID + A×x". আগে centroid, তারপর overturning moment, তারপর $P \propto A \cdot x$.

6.6 Cantilever Method - Reliable Workflow

19. At the top storey or selected storey cut, compute the centroid of all column areas.
20. Compute the overturning moment M_o of all lateral loads above the cut about the storey level.
21. Use $P_i = M_o A_i x_i / \sum(Ax^2)$ to obtain axial force in each column; keep x_i signed.
22. Move downward to the next storey and repeat using all lateral loads above that level.
23. Use joint and member equilibrium with the assumed mid-height / mid-span inflection points to obtain beam shear and end moments.
24. Tabulate all column P , V , M before drawing the final SFD/BMD.

6.7 Fast Symmetry Check

Three equal-area columns at $x = -b, 0, +b$ হলে $\Sigma Ax^2 = 2Ab^2$. তাই outer axial forces are equal and opposite: $P_{\text{left}} = -M_o/(2b)$, $P_{\text{mid}} = 0$, $P_{\text{right}} = +M_o/(2b)$. এটি sign/ratio check হিসেবে খুব useful.

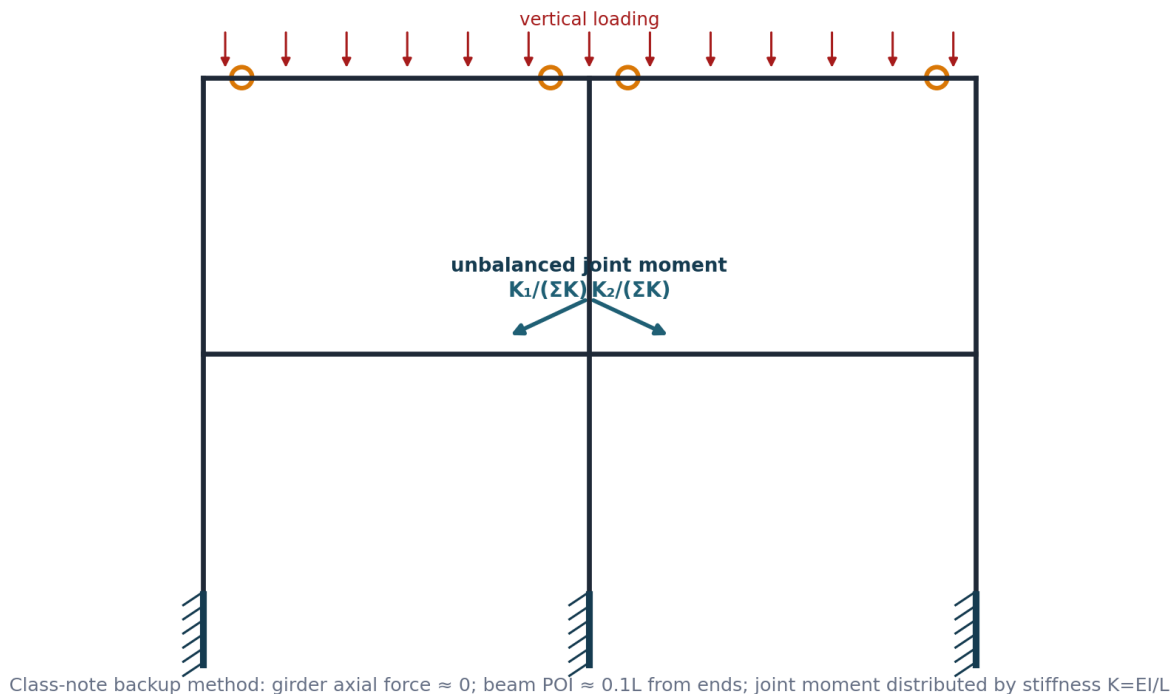
QB MATCH - Cantilever Cantilever-method frame appears repeatedly from 2016 onward and is paired with Portal in the modern 2022-2024 pattern. The solved QB file also repeatedly starts with centroid and $A \times x$ force ratios.

6.8 Portal vs Cantilever - Do Not Mix the Assumptions

Feature	Portal Method	Cantilever Method
Primary global action	Storey shear	Overturning moment
Main distribution assumption	Interior column shear ≈ 2 exterior shear	Column axial force $\propto A \times \text{distance from centroid}$
Inflection points	Mid-height columns + mid-span beams	Mid-height columns + mid-span beams
First calculation	Storey shear	Column-group centroid + overturning moment
Best memory cue	$I = 2E$	$P \propto A \cdot x$

Standard interpretation Portal and cantilever are different approximations; neither is universally “more accurate” for every frame. If the exam asks comparison, explain the different assumed resisting mechanisms first. A common textbook interpretation is portal action for lower/shear-dominant frames and cantilever action for taller/overturning-dominant frames.

6.9 Factor / Approximate Vertical-Load Method - Historical Backup



This compact topic follows the assumptions explicitly written in the Ramjan class note.

- Axial force in the girder/beam is taken as zero in the approximate idealization.
- Point of inflection in a girder/beam is assumed at approximately one-tenth of the span from each end.
- Unbalanced bending moment at a joint is distributed among connected columns according to their relative stiffness.

Joint-moment distribution idea

Bending stiffness factor: $K = EI / L$

Moment share of member $i \approx M_{\text{unbalanced}} \times K_i / \Sigma K$

25. Calculate the beam/girder end moments produced by the vertical loading under the adopted approximation.
26. At each joint, distribute the unbalanced moment to the columns in proportion to $K=EI/L$; a longer/weaker column carries less moment.
27. Use column equilibrium to obtain shear from the algebraic end moments and column height.
28. Draw the member SFD/BMD and verify joint moment equilibrium.

QB MATCH - Factor Historically visible especially in 2017-2019. Recent papers emphasize Portal + Cantilever, so Factor is a compact backup rather than the first study priority.

6.10 Matching Questions from the Question Bank

Year	Matching question pattern (paraphrased)
2016	Portal/cantilever assumptions + full cantilever-method building-frame analysis.
2017	Cantilever assumptions/analysis; compare approximate methods + factor-method analysis.
2018	Cantilever frame + factor-method frame.
2019	Cantilever frame; portal assumptions + factor method.
2020*	Shortened scan is incomplete for this block.
2021	Portal/cantilever assumptions + cantilever frame; separate portal-method frame.
2022	Cantilever method + portal method paired.
2023	Portal method + cantilever method paired in the recent standard form.
2024	Portal method + cantilever method paired again.
2024 BL	Portal method + cantilever method, same modern pattern.

Set 6 - Most Common Errors

- Using portal 2:1 column-shear rule inside a cantilever solution.
- Forgetting to recompute storey shear / overturning moment as you move downward.
- Using geometric center instead of area-weighted centroid when column areas differ.
- Dropping the sign of x_i in cantilever axial-force calculation.
- Drawing SFD/BMD before completing a member-end force table.
- Using the factor-method $0.1L$ beam inflection assumption in the portal/cantilever method.

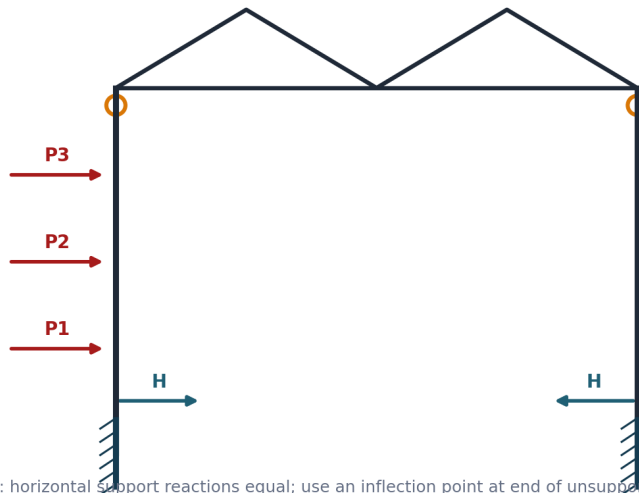
SET 7 - Mill Bent / Portal Application + Space Truss

QB STATUS Space truss is the more stable annual core. Mill-bent / industrial-frame portal application recurs beside it, especially in recent papers.

7A. Mill Bent / Bridge Portal Approximation

Mill bent সাধারণত industrial building-এর transverse frame/column-truss system। Uploaded class note-এ bridge portal / mill bent-এর জন্য building-frame portal method-এর থেকে আলাদা assumptions দেওয়া হয়েছে।

MILL BENT / BRIDGE PORTAL APPROXIMATION



Class-note assumptions: horizontal support reactions equal; use an inflection point at end of unsupported height of each fixed-base column

Keep this method separate from the ordinary building-portal 2:1 shear assumption.

7A.1 Class-Note Assumptions

- Horizontal support reactions are assumed equal.
- A point of inflection is assumed at the end of the unsupported height of each fixed-base column.

Critical distinction Set 6 building Portal → interior column shear = 2 exterior. Set 7 mill bent → use the special assumptions stated for the mill-bent problem. Mixing these two is a common mistake.

7A.2 Mill-Bent Solution Workflow

29. Draw the complete frame/truss-column FBD and calculate the global vertical reactions by equilibrium.
30. Apply the stated mill-bent assumption for horizontal base reactions.
31. Isolate each leg/column. Include all lateral distributed/point loads and the actions transferred by the roof/truss.
32. Use the assumed point of inflection to impose $M=0$ at the stated location where applicable.
33. Calculate leg shear and bending moment ordinates from equilibrium.
34. Draw separate SFD/BMD for the requested legs/girders only; label units and signs clearly.

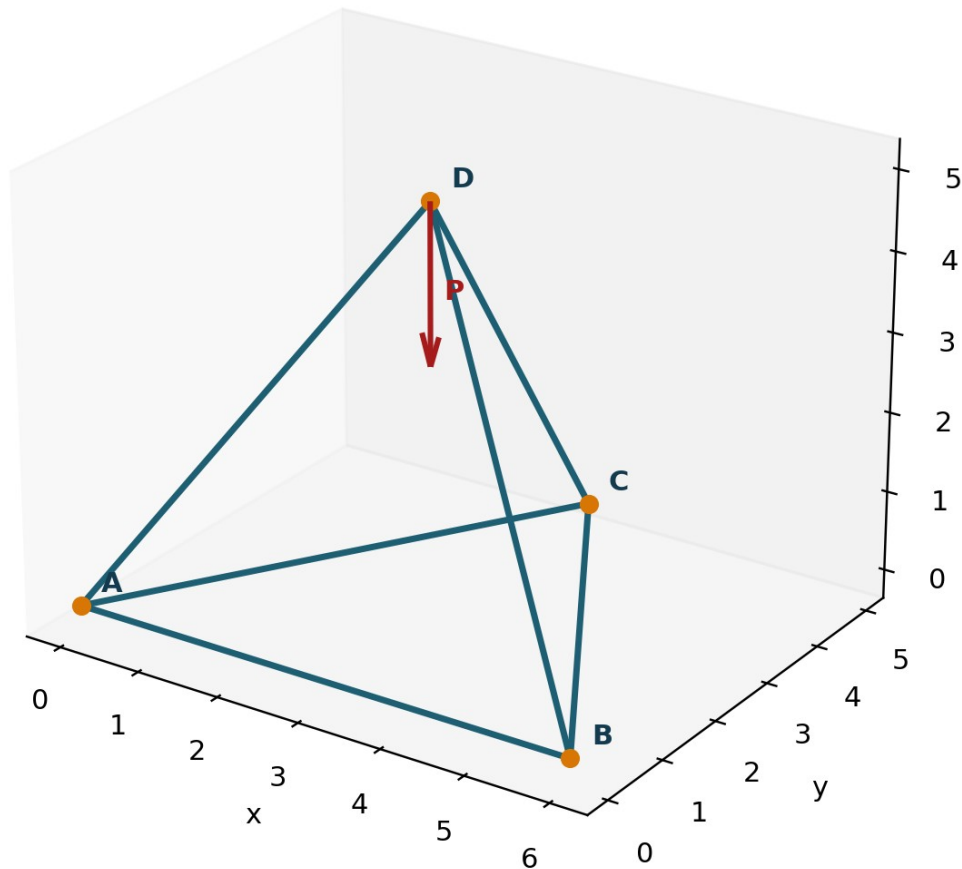
QB MATCH - Mill Bent Mill-bent / leg SFD-BMD appears in 2018, 2019 and becomes strongly visible again in 2023, 2024 and 2024 Backlog, usually beside portal/space-truss work.

7B. Space Truss

7B.1 Definition

Space truss হলো three-dimensional truss system যেখানে members universal/pin-type joints দিয়ে এমনভাবে connected যে ideal member মূলত axial force বহন করে। Plane truss-এর মতো শুধু ΣF_x ও ΣF_y যথেষ্ট নয়; এখানে x-y-z তিন direction-এর equilibrium লাগে।

Space truss: use 3D direction cosines at every member



Never solve a space-truss member from a 2D sketch alone; use the plan + elevation / coordinates together.

3D rigid-body equilibrium

$$\begin{aligned}\Sigma F_x &= 0, \quad \Sigma F_y = 0, \quad \Sigma F_z = 0 \\ \Sigma M_x &= 0, \quad \Sigma M_y = 0, \quad \Sigma M_z = 0\end{aligned}$$

7B.2 Methods Listed in the Class Note

Method	Use
Projection of forces on axes	Resolve all member/reaction forces into x, y, z components.
Projection of forces on a plane	Useful when a projected geometry simplifies one or more unknowns.
Tension-coefficient method	Use member force/length coefficient to write coordinate-form equilibrium.
Moment about an axis	Eliminate several unknown force components in the whole-truss reaction calculation.

7B.3 Direction Cosines - The Most Important Calculation

Member i-j-এর endpoints যদি (x_i, y_i, z_i) এবং (x_j, y_j, z_j) হয়, প্রথমে direction vector এবং length বের করো। তারপর member force-কে x, y, z component-এ resolve করো।

Direction-cosine relations

$$\begin{aligned}\Delta x &= x_j - x_i, \quad \Delta y = y_j - y_i, \quad \Delta z = z_j - z_i \\ L_{ij} &= \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} \\ l &= \Delta x/L, \quad m = \Delta y/L, \quad n = \Delta z/L \\ F_x &= F l, \quad F_y = F m, \quad F_z = F n\end{aligned}$$

Tiny check If $\Delta x:\Delta y:\Delta z = 3:4:12$, then $L=13$ and direction cosines = $3/13, 4/13, 12/13$. Their squares must add to 1.

7B.4 Tension Coefficient Method

Class note-এ tension coefficient method-এর কথা আছে। Member ij-এর tension coefficient $t_{ij} = F_{ij}/L_{ij}$ নিলে components সরাসরি coordinate difference দিয়ে লেখা যায়।

Tension coefficient

$$t_{ij} = F_{ij} / L_{ij}$$

At joint i: component vector = $t_{ij} (\Delta x, \Delta y, \Delta z)$

Sign convention Unknown member force first tension ধরে joint থেকে away arrow দাও। Solved value positive → tension; negative → compression.

7B.5 Determinacy / Stability Discussion

Space truss-এর exam theory-তে member-joint count একটি first check; কিন্তু 3D geometry এবং support arrangement অবশ্যই inspect করতে হবে। Standard perfect-space-truss count is $m + r = 3j$ for a stable determinate arrangement. $m+r < 3j$ সাধারণত insufficient/unstable and $m+r > 3j$ indicates static indeterminacy, provided the geometry itself is stable.

Geometry caution Count ঠিক হলেও coplanar/poorly arranged members বা inadequate 3D restraints mechanism তৈরি করতে পারে। তাই count + geometry দুটোই লিখবে।

7B.6 Space Truss - Exam Workflow

35. Choose and draw x-y-z axes exactly as in the plan/elevation.
36. Assign coordinates to every joint needed in the solution.
37. Solve external support reactions from the whole-space-truss FBD using force and moment equilibrium; moment about a convenient axis can eliminate unknowns.
38. Select a joint having at most three unknown member forces if possible.
39. For each connected member, calculate direction cosines or use tension coefficients.
40. Apply $\Sigma F_x=0, \Sigma F_y=0, \Sigma F_z=0$ at the joint.
41. Continue joint to joint; record T/C after every solved force and finish with a member-force table.

Common mistake	Correction
Using angle from elevation only	A member can be inclined in both plan and elevation; use full 3D coordinates.
Swapping x and y from plan	Write the coordinate table before equations.
Dropping negative sign	Negative under assumed tension means compression; do not change magnitude silently.
Trying >3 unknowns at a joint	Find another joint or solve reactions/other members first.

7.1 Matching Questions from the Question Bank

Year	Matching question pattern (paraphrased)
2016	Full space-truss support reactions and member forces.
2017	Space-truss numerical again.
2018	Define space truss / methods + solve; separate mill-bent leg SFD/BMD.
2019	Space truss + mill-bent SFD/BMD.
2020*	Shortened scan is incomplete for this block.
2021	Full space-truss support-reaction/member-force problem.
2022	Define determinacy/stability of space truss + full 3D solution.
2023	Space truss + mill-bent portal-method application.
2024	Mill-bent portal application + space-truss numerical.
2024 BL	Mill-bent column diagrams + space-truss reactions/member forces.

SET 8 - Three-Moment Equation + Suspension Cable

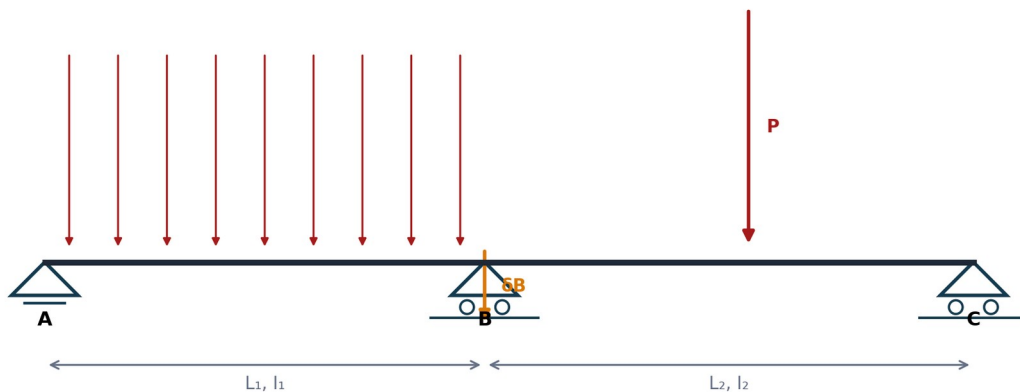
QB STATUS One of the most persistent and formulaic Section B sets. Continuous beam by three-moment equation + a cable tension/dip/length problem appears year after year.

8A. Three-Moment Equation

8A.1 Why It Is Used

Continuous beam statically indeterminate হওয়ায় শুধু equilibrium দিয়ে support moments পাওয়া যায় না। Three-Moment Equation তিন consecutive supports A-B-C-এর support moments M_A , M_B , M_C -কে দুই adjacent span-এর load, EI এবং support settlement-এর সাথে relate করে।

THREE CONSECUTIVE SUPPORTS → ONE THREE-MOMENT EQUATION



The equation is applied to three consecutive supports at a time; then each span is solved separately by statics.

Memory trick "3 supports = 2 spans = 1 equation". Unknown support moments যত বেশি, overlapping triples A-B-C, B-C-D, C-D-E ... ব্যবহার করো।

8A.2 General Area Form - Cleanest and Safest

The uploaded class/QB solution uses the expanded P-k-w-settlement form. নিচের area form একই method-কে compactভাবে লিখে, এবং point load/UDL/combined loading-এ safer because you can obtain the simply-supported BMD area directly.

Three-moment equation (support settlement δ taken positive downward)

$$\begin{aligned} M_A L_1/I_1 + 2M_B(L_1/I_1 + L_2/I_2) + M_C L_2/I_2 \\ = -6[A_1 \bar{x}_1/(L_1 I_1) + A_2 \bar{b}_2/(L_2 I_2)] \\ + 6E[(\delta_B - \delta_A)/L_1 + (\delta_B - \delta_C)/L_2] \end{aligned}$$

A_1 , A_2 = area of the simply-supported bending-moment diagrams caused by external loads on spans AB and BC. $\bar{x}_1$ = centroid distance of the AB free-BMD area from A; $\bar{b}_2$ = centroid distance of the BC free-BMD area from C. I_1 , I_2 are the span second moments of area. If there is no settlement, the δ -term is zero.

Consistency note If you use the expanded class-note formula instead of this area form, keep the exact same moment and settlement sign convention throughout. Never mix pieces from two sign conventions.

8A.3 Constant EI / No-Settlement Reduction

When I is constant and all supports are at the same level

$$M_A L_1 + 2M_B(L_1+L_2) + M_C L_2 = -6[A_1 \bar{x}_1/L_1 + A_2 \bar{b}_2/L_2]$$

8A.4 Very Useful Free-BMD Shortcut

Load on a simple span	Free-BMD fact	Three-moment use
Full-span UDL w	BMD is parabola; area $A = wL^3/12$; centroid at $L/2$	$A\bar{x}/L = wL^3/24$, so the -6 term becomes $-wL^3/4$ for that span.
Single point load	Free BMD is two triangles meeting below the load	Draw the two triangles; calculate total area and centroid - safer than memorizing a special k-form.
Combined loads	Superpose simple-support BMDs	Add areas/first moments algebraically before substitution.

8A.5 Sanity Example: Equal Two-Span Beam with UDL

A-B-C, both spans = L, constant EI, simple end supports $M_A=M_C=0$, same UDL w on both spans. Each span has $A=wL^3/12$ and centroid $L/2$. Three-moment equation gives $4M_B L = -wL^3/2$, hence:

Classical check value

$$M_B = -wL^2/8$$

Why useful If your equal-span/equal-UDL calculation gives a different middle support moment, recheck the load-area term and sign.

8A.6 Full Numerical Workflow

42. Make a top table: support moments known/unknown, span L, I, loads, and support settlements δ .
43. For every adjacent pair of spans, calculate the free simply-supported BMD area and centroid (or use the class P-k-w form).
44. Write one three-moment equation for A-B-C, then B-C-D, etc.
45. Use known conditions: simple support moment = 0; internal hinge moment = 0; a cantilever/overhang may provide a known support moment from local equilibrium.
46. Solve the simultaneous equations for unknown support moments.
47. Now treat each span separately with its two end moments + external loads and obtain end reactions by statics.
48. At an internal support, total vertical reaction is the sum of the contributions from its left and right spans.
49. Draw SFD, then BMD. The BMD ordinates at supports must equal the solved support moments.

8A.7 Support Settlement and Variable I

QB-তে settlement এবং different I খুব common। তাই প্রথম line-এই I এবং δ table বানানো safest. Settlement positive downward ধরে area-form-এর settlement contribution central support B-এর relative settlement দিয়ে লেখা হয়েছে:

Use signed settlements consistently

$$\text{Settlement term} = 6E[(\delta_B - \delta_A)/L_1 + (\delta_B - \delta_C)/L_2]$$

Shortcut All supports same level $\rightarrow$ settlement term immediately 0. $I_1=I_2 \rightarrow$ common I may cancel after multiplying the equation by I.

Situation	Fast handling
Simple end support	Support moment = 0
Internal hinge	Moment at hinge = 0
Different I	Keep L/I terms separately; do not cancel I too early
Support settlement	Assign signed δ to every support first

Overhang/cantilever at an end	Find the known support end moment from that overhang/cantilever by local equilibrium before the three-moment equations
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QB MATCH - Three-Moment Asked throughout 2016-2024. Variations include support settlement, variable I, end overhang/cantilever and complete SFD/BMD after the support moments are found.

8B. Suspension Cable

8B.1 Physical Idea

Cable খুব flexible; ideal analysis-এ এটি bending moment বা compression resist করে না, শুধু tension বহন করে। Loading-এর ধরন অনুযায়ী cable-এর shape পরিবর্তিত হয়। Suspension bridge এবং transmission line-এর মতো systems-এ এই behavior ব্যবহার করা হয়।

Cable fact	Exam meaning
Horizontal component H	Only vertical loads থাকলে H is constant throughout the cable.
Vertical reaction	Same vertical loading-এর equivalent simply-supported beam reaction-এর মতো পাওয়া যায়.
Moment of cable segment	Any cut segment satisfies ordinary equilibrium; ideal cable itself carries no bending moment.
Tension	$T = \sqrt{(H^2 + V^2)}$ for the local vertical component V.
Lowest point	Cable tangent is horizontal $\rightarrow V=0 \rightarrow T=H$ at that point.

Memory trick "Beam gives V, cable adds H." আগে vertical reactions exactly beam-এর মতো বের করো; তারপর dip/geometry থেকে H; শেষে $T = \sqrt{(H^2 + V^2)}$.

8B.2 Same-Level Supports + Concentrated Loads

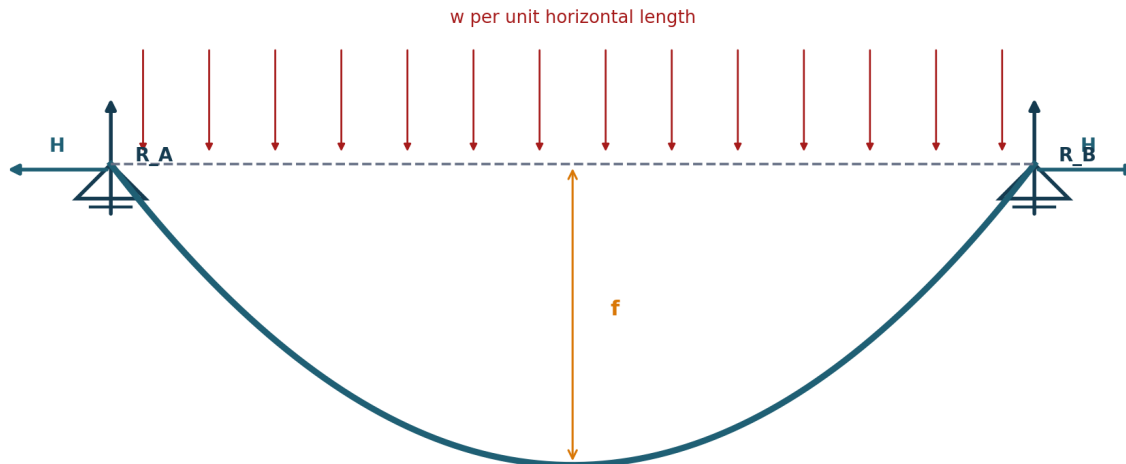
- Find R_A and R_B from $\Sigma V=0$ and $\Sigma M=0$ exactly as for a simply-supported beam under the same vertical loads.
- Use a point where the cable ordinate/dip y is known. Taking moment of the cable segment about that point gives H.
- Equivalently for same-level supports, $M_{\text{beam}}(x) = H y(x)$ when y is measured from the support chord.
- After H is known, determine vertical component V in each cable segment from vertical equilibrium.
- Compute segment tension $T=\sqrt{(H^2+V^2)}$; the largest value commonly occurs at the support having the larger |vertical component|.

Same-level cable shortcut at a point with known ordinate

$$H = M_{\text{beam}}(x) / y(x)$$

QB pattern Concentrated-load cable problems often give one or more dips/ordinates and ask maximum tension and cable length. Use segment-by-segment straight-line geometry only for point-load cable portions.

8B.3 Same-Level Supports under UDL - Must Memorize



$$H = wL^2/(8f) \quad T_{max} = \sqrt{H^2 + (wL/2)^2} \quad S \approx L + 8f^2/(3L)$$

UDL is taken per unit horizontal length. The cable is parabolic.

Same-level parabolic cable

$$\begin{aligned} R_A &= R_B = wL/2 \\ H &= wL^2/(8f) \\ y(x) &= wx(L-x)/(2H) = 4fx(L-x)/L^2 \\ T_{max} &= \sqrt{H^2 + (wL/2)^2} \\ \text{Cable length } S &\approx L + 8f^2/(3L) \end{aligned}$$

Here L = horizontal span, f = maximum dip at midspan, and y is measured downward from the straight chord joining the supports. Maximum tension occurs at the supports because $|V|$ is maximum there.

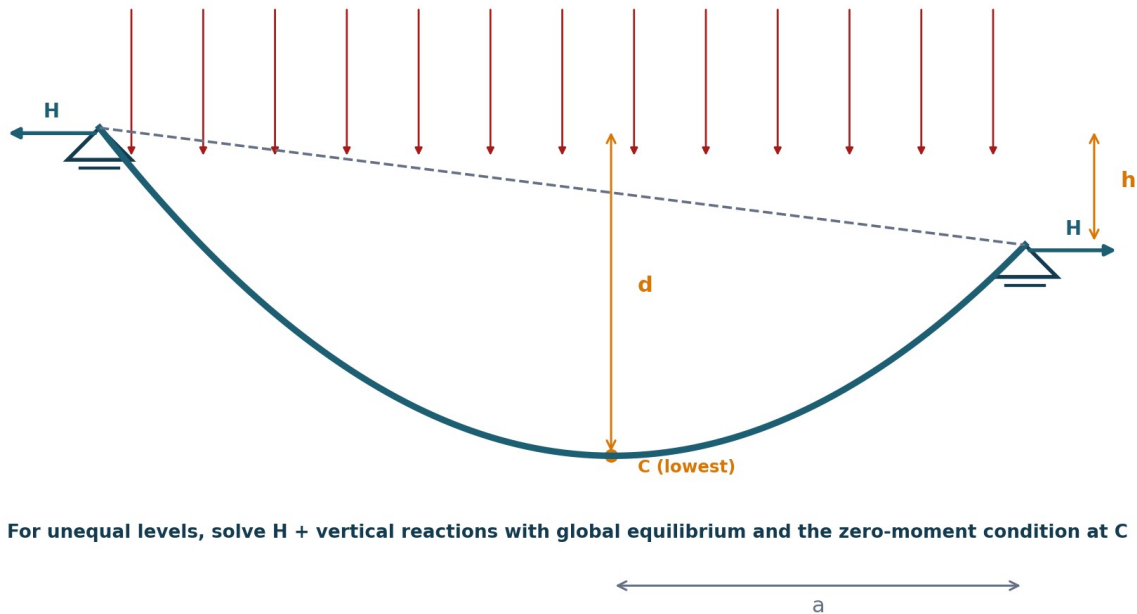
Units check w [force/length], L^2/f [length] $\rightarrow H$ has force units. If H comes out as force $\times$ length, a length factor is missing.

8B.4 Shape Relation under UDL

UDL per horizontal length হলে cable-এর parabola directly moment relation থেকে আসে: simply-supported beam moment $M(x) = wx(L-x)/2$ and ideal cable equilibrium gives $M = Hy$. Therefore $y = wx(L-x)/(2H)$. At $x = L/2$, $y = f \rightarrow H = wL^2/(8f)$.

Derivation memory Beam moment shape under full UDL is parabolic $\rightarrow$ cable under horizontal UDL becomes the matching funicular parabola.

8B.5 Supports at Different Levels under UDL



A is h above B ; C is the lowest point. In unequal-level problems, use equilibrium instead of forcing the symmetric formula.

Class-note/textbook setup: horizontal span = L , support A is h above support B , lowest point C is at horizontal distance a from B and d below A level. Then:

Four core relations for the unequal-level UDL cable

$$\begin{aligned} H_A &= H_B = H \\ R_A + R_B &= wL \\ R_A L - wL^2/2 - Hh &= 0 \\ \text{At } C: R_A(L-a) - Hd - w(L-a)^2/2 &= 0 \end{aligned}$$

Lowest-point shortcut If x is the horizontal distance from A to the lowest point C , the vertical component at C is zero. For UDL only: $R_A = wx$ and $Hd = wx^2/2$ (where d is C below A level). Combine these with global equilibrium to solve x and H .

8B.6 Cable Length when the Lowest Point is not at Midspan

Lowest point C যদি span-কে x এবং $L-x$ horizontal projections-এ ভাগ করে, এবং C থেকে support A/B -এর vertical rises যথাক্রমে f_A এবং f_B হয়, small-slope parabolic approximation ব্যবহার করলে:

Unequal-side parabolic length approximation

$$S \approx x + 2f_A^2/(3x) + (L-x) + 2f_B^2/[3(L-x)]$$

Symmetric case-এ $x=L/2$ and $f_A=f_B=f$ দিলে equationটি $S \approx L + 8f^2/(3L)$ -এ reduce করে।

8B.7 Maximum Tension

Support tension check

$$\begin{aligned} T_A &= \sqrt{H^2 + R_A^2} \\ T_B &= \sqrt{H^2 + R_B^2} \\ T_{\max} &= \max(T_A, T_B) \end{aligned}$$

Exam shortcut UDL unequal-level case-এ larger vertical reaction-এর support tension usually larger. কিন্তু final answer দেওয়ার আগে T_A এবং T_B দুটোই compare করো।

8B.8 Cable under Point Loads - Length and Segment Forces

Concentrated loads থাকলে cable polygonal straight segments-এর series হয় (loads-এর মাঝখানে distributed load না থাকলে). H constant থাকে, কিন্তু vertical component load point পার হলে jump করে।

Task	Fast method
Vertical reactions	Whole cable equilibrium = simply-supported beam reaction calculation.
Horizontal H	Use zero-moment equilibrium at a point with known dip/ordinate.
Vertical component of each segment	Start from a support reaction; subtract/add point loads as you move across.
Segment tension	$T_i = \sqrt{(H^2 + V_i^2)}$.
Cable length	Get vertical ordinate differences from $H y = M_{\text{beam}}$ at load points; each load-to-load piece is straight, so length = $\sqrt{(\Delta x^2 + \Delta y^2)}$. Sum all segments.

Very useful relation For same-level support cables under vertical point loads: at any load/joint ordinate y_i , $H y_i = M_{\text{beam}}(x_i)$. This converts a beam BMD ordinate directly to cable geometry.

8.1 Matching Questions from the Question Bank

Year	Matching question pattern (paraphrased)
2016	Three-moment with settlement + diagrams; cable maximum tension plus backstay/saddle tension.
2017	Three-moment continuous beam; suspension-bridge cable maximum tension.
2018	Three-moment beam with variable I; cable maximum tension + cable length.
2019	Three-moment with settlement/variable I; unequal-support cable max tension, dip and length.
2020*	Three-moment equation with support settlement visible.
2021	Three-moment beam + unequal-level cable maximum tension.
2022	Three-moment beam + cable-supported roof under UDL.
2023	Three-moment beam + unequal-level UDL cable maximum tension.
2024	Three-moment beam + suspension cable maximum tension.
2024 BL	Three-moment beam with several settlements/variable I + unequal-level cable maximum tension.

Section B - High-Frequency Theory Questions

Priority	Question / answer focus
Very High	Define influence line. Explain the equilibrium method and the difference between ILD and SFD/BMD.
Very High	Why is the influence line straight/piecewise straight for a statically determinate structure?
High	State the importance/applications of influence line diagrams.
High	State and explain Müller-Breslau principle.
Very High	Write the assumptions of Portal method and Cantilever method.
High	Differentiate Portal and Cantilever methods.
Medium	State the assumptions of the class-note factor / approximate vertical-load method.
High	Define a space truss. State the methods of analysis.
High	Discuss determinacy and stability of a space truss.
High	State the main assumptions used in approximate analysis of bridge portal / mill bent.
High	State the purpose of the Three-Moment Equation and the meaning of its terms.
High	Explain why the horizontal component of cable tension is constant under vertical loading.
High	Derive / state $H=wL^2/(8f)$ and the parabolic cable equation for a same-level UDL cable.
Medium	Explain why maximum cable tension occurs at a support for the symmetric UDL case.

Section B - Proof / Derivation / Formula-Logic List

Item	What you should be able to show
ILD by equilibrium	Derive reaction/shear/moment as piecewise functions of moving unit-load position x .
Müller-Breslau	Release the response restraint and identify the positive kinematic shape.
Portal method	From storey shear + 2:1 column-shear assumption, obtain column end moments using mid-height POI.
Cantilever method	From $P_i = C A_i x_i$ and $\Sigma(P_i x_i) = M_o$ derive $P_i = M_o A_i x_i / \Sigma(Ax^2)$.
Space-truss direction cosines	From endpoint coordinates derive L, l, m, n and component equations.
Three-moment UDL load term	Use free-BMD area $A=wL^3/12$ and centroid $L/2$ to obtain the $-wL^3/4$ contribution for one UDL span.
Cable same-level UDL	From $M_{\text{beam}} = Hy$ derive $y = wx(L-x)/(2H)$, then $H = wL^2/(8f)$.
Cable length	Use $ds \approx [1 + 0.5(dy/dx)^2]dx$ for a shallow parabola to obtain $S \approx L + 8f^2/(3L)$.

Priority advice For exam preparation, derivation logic matters more than memorizing one old numerical. If the geometry changes, the same equilibrium backbone still works.

Section B - Final Formula Sheet

Influence Line

$$\begin{aligned} \text{Simple beam: } R_A &= (L-x)/L, \quad R_B = x/L \\ \text{At C, a from A: } V_C &= -x/L \quad (x < a); \quad V_C = (L-x)/L \quad (x > a) \\ M_C &= xb/L \quad (x < a); \quad M_C = a(L-x)/L \quad (x > a) \\ \text{Shear ILD jump at C} &= +1 \end{aligned}$$

Portal Method

$$\begin{aligned} V_{\text{storey}} &= \Sigma \text{ lateral loads above} \\ \text{Typical equal frame: } V_{\text{interior}} &= 2 V_{\text{exterior}} \\ \text{Column end moment magnitude} &= V_{\text{column}} h/2 \end{aligned}$$

Cantilever Method

$$\begin{aligned} \bar{x} &= \Sigma Ax / \Sigma A \\ P_i &= M_o A_i x_i / \Sigma (A_j x_j^2) \end{aligned}$$

Space Truss

$$\begin{aligned} L &= \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} \\ l &= \Delta x/L, \quad m = \Delta y/L, \quad n = \Delta z/L \\ \text{Joint: } \Sigma F_x &= \Sigma F_y = \Sigma F_z = 0 \\ \text{Tension coefficient } t &= F/L \end{aligned}$$

Three-Moment

$$\begin{aligned} &M_A L_1/I_1 + 2M_B(L_1/I_1 + L_2/I_2) + M_C L_2/I_2 \\ &= -6[A_1 \bar{x}_1/(L_1 I_1) + A_2 \bar{b}_2/(L_2 I_2)] + 6E[(\delta_B - \delta_A)/L_1 + (\delta_B - \delta_C)/L_2] \end{aligned}$$

Cable

$$\begin{aligned} \text{Vertical reactions: solve exactly like simply-supported beam} \\ \text{Same level: } M_{\text{beam}}(x) &= H y(x) \\ \text{UDL same level: } H &= wL^2/(8f) \\ y &= 4f x(L-x)/L^2 \\ T &= \sqrt{(H^2 + V^2)}; \quad T_{\text{max support check}} \\ S &\approx L + 8f^2/(3L) \end{aligned}$$

Final 3-of-4 Checklist - Section B

Exam hall-এ set select করার আগে এই checklist mentally run করো:

Set	You are exam-ready if...
SET 5 - ILD	You can define x-ranges, derive reactions/shear/moment piecewise, handle internal hinge, and draw both beam + frame ILD.
SET 6 - Approximate Frame	You can state portal/cantilever assumptions, calculate storey shear/centroid, distribute forces correctly, and complete member SFD/BMD.
SET 7 - Mill Bent + Space Truss	You can keep mill-bent assumptions separate, solve 3D reactions, compute direction cosines and member T/C.
SET 8 - Three-Moment + Cable	You can handle settlement + variable I, solve support moments then reactions/diagrams, and calculate cable H, dip, length and maximum tension.

Default answer route Set 5 + Set 6 + Set 8 is the default high-confidence route from the Question Bank analysis. Use Set 7 as a strong switch if space truss is one of your better topics.

Last-Minute Error Checklist

- ILD: write x-ranges before equations.
- Portal vs Cantilever: never mix 2:1 shear with $A \times x$ axial distribution.
- Mill bent: use its own stated assumptions.
- Space truss: make coordinate table; direction-cosine squares must sum to 1.
- Three-moment: make L-I-load-settlement table before substitution.
- Cable: keep H separate from vertical reaction; at lowest point vertical component is zero.
- All diagrams: label sign, unit and key ordinates.

Prepared from the uploaded CE 3111 course materials and Question Bank set distribution.

Made by 2223033